

Gabrielle Johnson's Week 7 Interim Benchmark Table

Preamble

Evaluation prioritises per class recall for ESI levels 2–4 because it shows how reliably the model identifies patients at each ESI level and conversely, how many cases it misses.

Precision is assessed as a secondary safeguard because a model that achieves high recall by routinely assigning patients to more urgent categories would generate an unsustainable number of false alarms, increasing staff workload and ultimately undermining trust.

Class specific under triage rates are also monitored. This approach follows consultation with Dr. De Freitas, who noted that ESI 1 patients are generally clinically apparent at presentation and that the model's greatest value lies in distinguishing the more ambiguous middle levels. The central concern is therefore ensuring that patients in ESI levels 2–4 are not assigned a lower urgency than their condition warrants as this could be life threatening. Under triage rate captures this risk directly unlike recall, which treats misclassification in either direction equally, it isolates errors in the clinically dangerous (under triage) direction.

Benchmark Table

**UT-Under Triage*

Model	Accuracy	Recall ESI 2	Recall ESI 3	Recall ESI 4	Precision ESI 2	Precision ESI 3	Precision ESI 4	UT * ESI 2 (%)	UT* ESI 3 (%)	UT* ESI 4 (%)	Macro F1	Training Time (s)	Inference Time (ms)	Explainability
LogReg (base)	0.667	0.609	0.757	0.589	0.716	0.661	0.610	39.0	8.8	1.5	0.493	19.2	0.005	High
LogReg (+demo)	0.686	0.620	0.775	0.626	0.736	0.677	0.636	37.9	8.1	1.5	0.510	6.8	0.006	High
LogReg (+demo+FE)	0.690	0.625	0.779	0.630	0.744	0.681	0.636	37.4	8.3	1.5	0.520	15.1	0.003	High
LogReg weighted (+demo+FE)	0.668	0.785	0.630	0.630	0.631	0.717	0.638	21.5	8.2	1.5	0.481	16.7	0.003	High
Tree (base)	0.614	0.506	0.833	0.247	0.710	0.584	0.612	49.4	3.6	1.3	0.349	1.6	0.002	High
Tree (+demo)	0.630	0.540	0.778	0.438	0.695	0.617	0.569	46.0	7.5	1.1	0.391	2.1	0.002	High
Tree (+demo+FE)	0.627	0.544	0.770	0.435	0.677	0.615	0.581	45.6	7.2	1.1	0.394	2.3	0.001	High

Model	Accuracy	Recall ESI 2	Recall ESI 3	Recall ESI 4	Precision ESI 2	Precision ESI 3	Precision ESI 4	UT * ESI 2 (%)	UT* ESI 3 (%)	UT* ESI 4 (%)	Macro F1	Training Time (s)	Inference Time (ms)	Explainability
Neural Network	0.688	0.604	0.797	0.609	0.761	0.672	0.637	39.5	8.2	1.8	0.505	9.4	0.005	Low
Random Forest	0.649	0.572	0.805	0.413	0.711	0.626	0.640	42.8	5.1	1.5	0.394	58.3	0.108	Medium
HGB	0.692	0.652	0.774	0.602	0.741	0.687	0.639	34.6	8.0	2.6	0.519	51.9	0.156	Low
HGB (weighted)	0.636	0.797	0.620	0.430	0.593	0.682	0.679	19.6	4.9	2.9	0.415	8.4	0.020	Low